C Programming Intro: Project 1

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My project summary

I completed four small projects to practise C programming and basic Arduino work. Three programs run in the terminal, and the smart parking system runs in a Tinkercad simulation.

1. Water quality monitoring

I enter the water temperature and turbidity. The program calculates a water quality index and shows the result as Good, Warning, or Critical.

2. Mobile money transactions

I use a menu to deposit money, withdraw money, check the balance, and view a transaction summary. The program rejects invalid amounts and withdrawals above the balance.

3. Delivery route analysis

I enter delivery distances. The program shows the total, average, longest route, and how many routes are above a chosen limit. It also calculates the total using recursion.

4. Smart parking system

I use an ultrasonic sensor to detect an object. At 20 cm or less, the red light and buzzer turn on. Above 20 cm, the green light shows that the space is free.

More details

My full explanations, source code, screenshots, and circuit details are in my project repository:

https://github.com/KaberaSamuel/c-intro-project-1